

STATE LEVEL ROBOT MAKING COMPETITION

“ROBOFEST-2.0”

UNDER SCIENCE, TECHNOLOGY AND INNOVATION (STI) FUND

Proposal submitted on

“TYPE OF OUR ROBOT”

SUBMITTED BY

Student Team	Mentor
Himanshu Laddhad (U19ME107)	Dr. Harshit K. Dave
Atul Dhamija (U19ME108)	Associate Professor, MED
Naman Jain (U19ME160)	
Dhruv Oza (U19EC160)	
Prabhakar Jaiswal (U19EE073)	



S. V. National Institute of Technology, Surat 395007

GUJARAT COUNCIL ON SCIENCE AND TECHNOLOGY

Dept. of Science and Technology, Govt. of Gujarat

Application format for the

IDEATION STAGE FOR ROBOT MAKING COMPETITION

ROBOFEST, GUJARAT-2.0

1. Proposal submitted for Robot: (Please tick mark the category of robot proposed)
 1. Four-legged Robot with quadrupedal motion
 2. Underwater Robot or Submarine Robot
 3. Rovers (Eight wheels, 3 to 4 feet size with the camera mounted, auto memory/GPS guided.
 4. Prosthetic limbs with remote sensors.
 5. Pipe Climbing robots
 6. Powered Exoskeletons
 7. Microrobots
2. Type of Institution:
Government: ☐ Private-Aided: ☐ Private-Unaided ☐ Local Body: ☐
3. Name of Institute/Department/Organization: Sardar Vallabhbhai National Institute Of Technology
4. Complete Postal Address with Pin code: Ichchhanath Surat- Dumas Road, Keval Chowk, Surat, Gujarat 395007
5. Name of affiliated University/Board Name: Sardar Vallabhbhai National Institute Of Technology
6. Mentor/Coach/Faculty Guide Name: Dr. Harshit K. Dave
7. Designation of Mentor/Coach/Faculty Guide: Associate Professor
8. Experience of Mentor/Coach/Faculty in years: 14years
9. Email address: hkd@med.svnit.ac.in
10. Mobile Contact No: 8140434777
11. Office No: (0261)-2201988, 2204325
12. Proposed Team of Making robot: (Maximum five participant per team allowed)

Sr. No.	Name of Student/ Research scholar	Institute Name	Department	Programme enrolled	Year (I/II/III/IV)	Mobile No.
1.	Atul Dhamija	SVNIT	Mechanical	B.Tech	II	9119041566
2.	Prabhakar Jaiswal	SVNIT	Electrical	B.Tech	II	8709187059
3.	Naman Jain	SVNIT	Mechanical	B.Tech	II	9413751259
4.	Himanshu Laddhad	SVNIT	Mechanical	B.Tech	II	8888431372
5.	Dhruv Oza	SVNIT	ECE	B.Tech	II	9408972411

13. Capability of the Organization / Individual specific to Robot Making:

a. Expertise available with the Mentor/Coach/Faculty Guide: (Not more than 100 words)

Dr. Harshit K. Dave is involved in teaching and research activities related to advance manufacturing processes, additive manufacturing and automation. He has undertaken 02 research projects sponsored by DST, New Delhi and GUJCOST, Gandhinagar. He has over 115 research publications and has filed 05 patents. He has developed an advanced manufacturing laboratory at the department that houses some of the best state of the art machines and instruments like micro machining setup, EDM, FDM and SLA 3D printers and some of the software like Ansys, Catia, solidworks, hyper works suit, etc. Under his mentorship, a student team won Robofest – I completion under the category of prosthetic limb.

b. List of Participation in past in any Robot Making Competitions:

Event Name	Venue	Type of Competition (State/national/International)	Type of Robot submitted	Achievement
Mind bend	NIT Surat	National	RC Plane	2nd Runner up
Mind bend	NIT Surat	National	Line Follower	Participation

14. Name of the authority in whose name Cheque / Demand Draft should be drawn.

Name of the Account Holder: Research & Consultancy Account SVNIT Surat

Name of the Bank: State Bank of India

Bank Account No: 35373382053

IFSC Code: SBIN0003320

MICR Code: 395002012

UNDERTAKING

I **Dr. Harshit K. Dave** (Mentor/Guide Name) on behalf of my team authorized to give undertaking that on selection of our team at Level 1 (Ideation) stage, we assure and commit to participate in the subsequent levels and to submit Level 2 (Proof of Concept) and Level 3 (Proto type) robot as per guidelines of the GUJCOST without fail, otherwise GUJCOST will take necessary action to recover the fund if disbursed for any Stage.

Dr. Harshit K. Dave

Director

Stamp of Institute/University

TECHNICAL DETAILS FOR PROPOSED ROBOT

(Wherever necessary separate sheet/page is allowed to attach; Institute may submit extra details if find necessary)

1. Type of Robot: **Underwater Robot**

2. Robot Assembly Design (Proposed Diagram): Drawings each part of the robot is preferred as an attachment. (CAD drawings are preferred) -

For this please Refer Annexure- 1

3. Components to be used: (Enlist all the components with their make/company in four groups as enlisted in the following table:

a. List of Structure components: like beams, bushes, shafts, belts, plates, pins, pulleys, wheel, connectors, batteries, motors, etc.

For this please Refer Annexure- 2(a)

b. List of Motion Components: like Chain, sprockets, flaps, etc.

For this please Refer Annexure- 2(b)

c. List of Electronics Components: like Smart ports, switches, joysticks, controllers, LED/LCD screen, power supply, programming components, etc.

For this please Refer Annexure- 2(c)

d. List of Other Accessories: Clothes, plastic eyes/ear/feeling like real all external components which are for the look.

For this please Refer Annexure- 2(d)

4. The methodology of Making Robot: (Please write Entire Technical Specifications of Proposed Robot with brief notes and diagrams)

For this please Refer Annexure- 3

5. Application of proposed Robot in a societal context: (Not more than 100 words)

For this please Refer Annexure- 4

6. Size of robot proposed for Proof of concept (Small version):

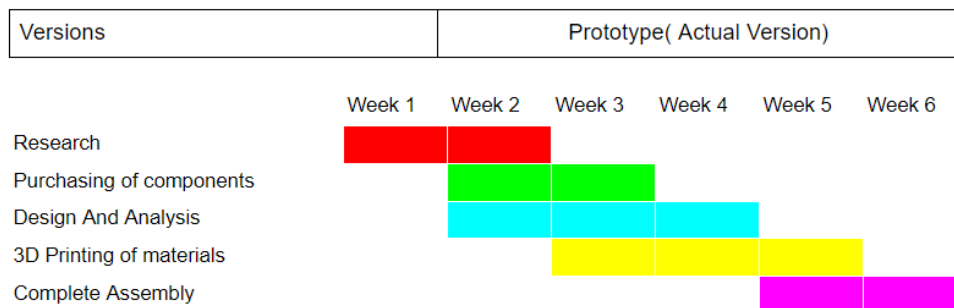
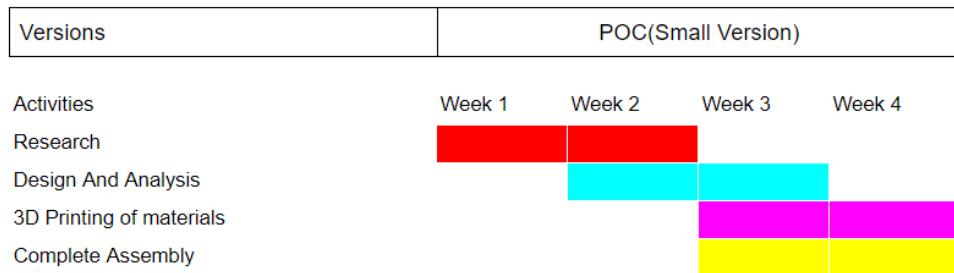
a. Length in cm- 30* cm b. Width in cm -30* cm c. Height in cm - 4* cm

7. Size of robot proposed as Proto type (Actual Version)

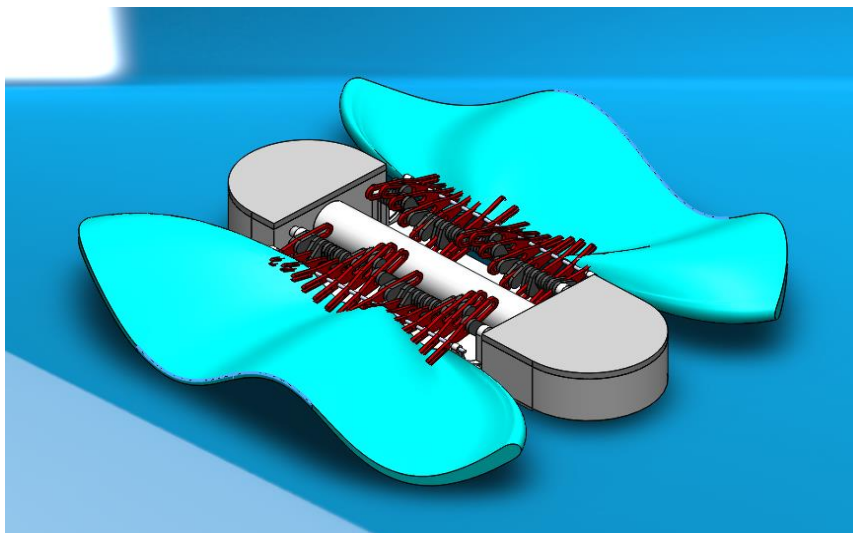
a. Length in cm- 70* cm b. Width in cm - 74* cm c. Height in cm - 9* cm

***Dimensions may vary according to final design and analysis**

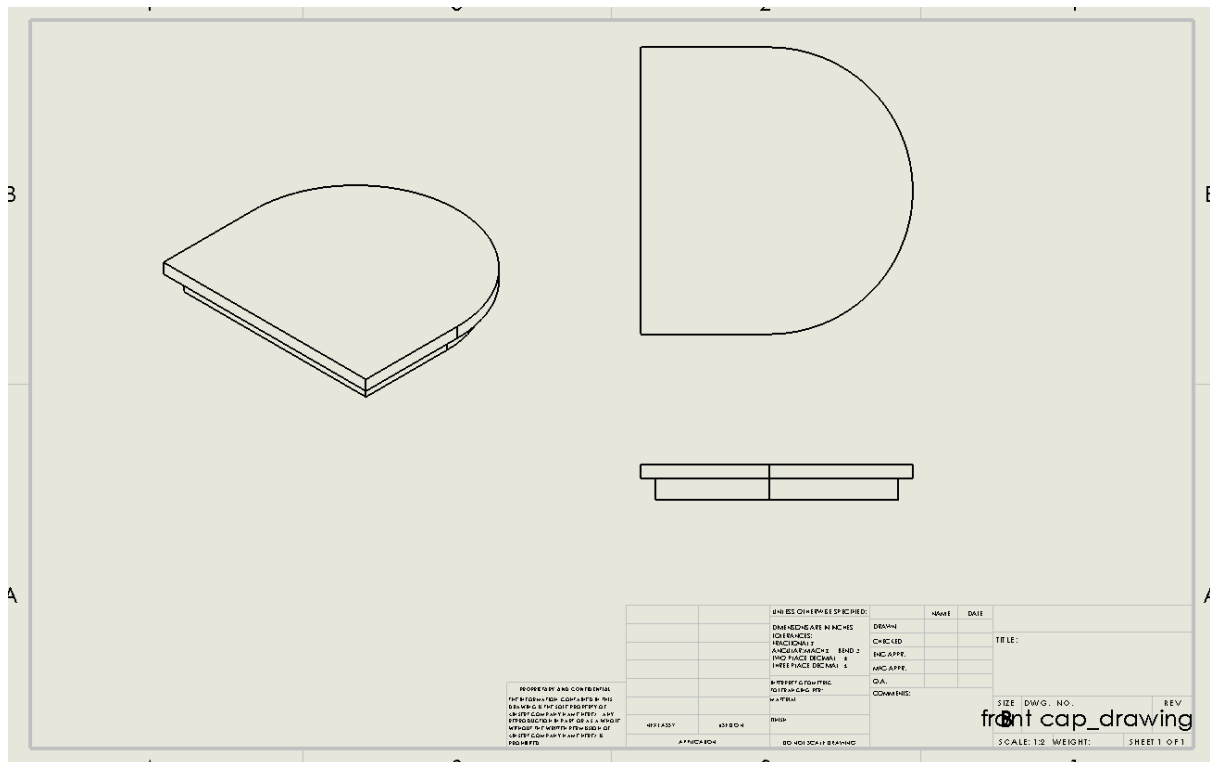
8. Timeline for Robot making with milestones. (Divided in activities vs. no. of days)

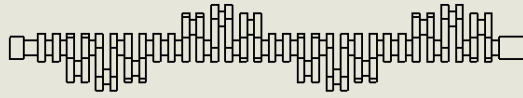


9. Please attach the proposed outline (photograph) for understanding of the evaluation committee.

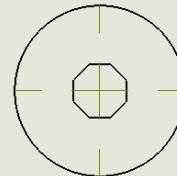
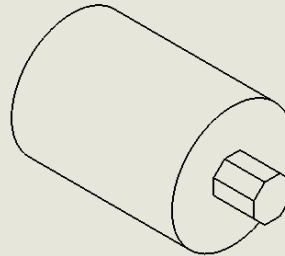


Robot Assembly Design (Proposed Diagram): Drawings each part of the robot are preferred as an attachment. (CAD drawings are preferred)

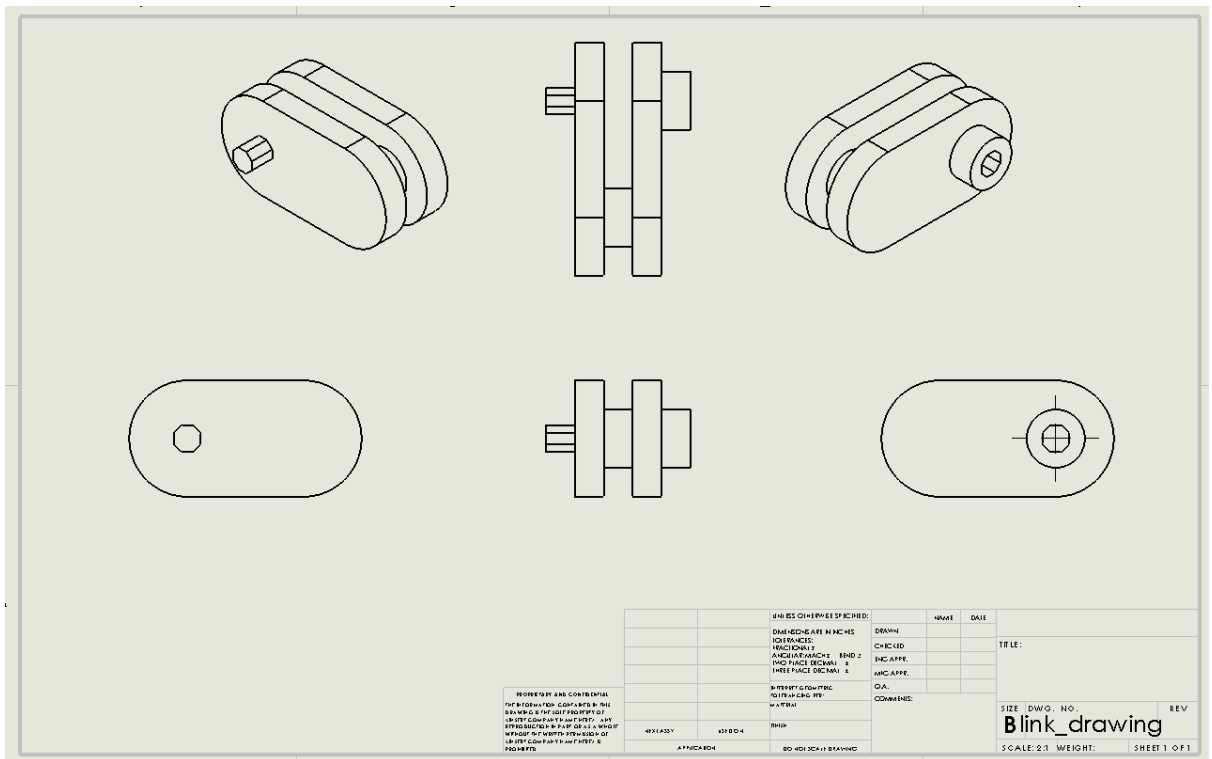
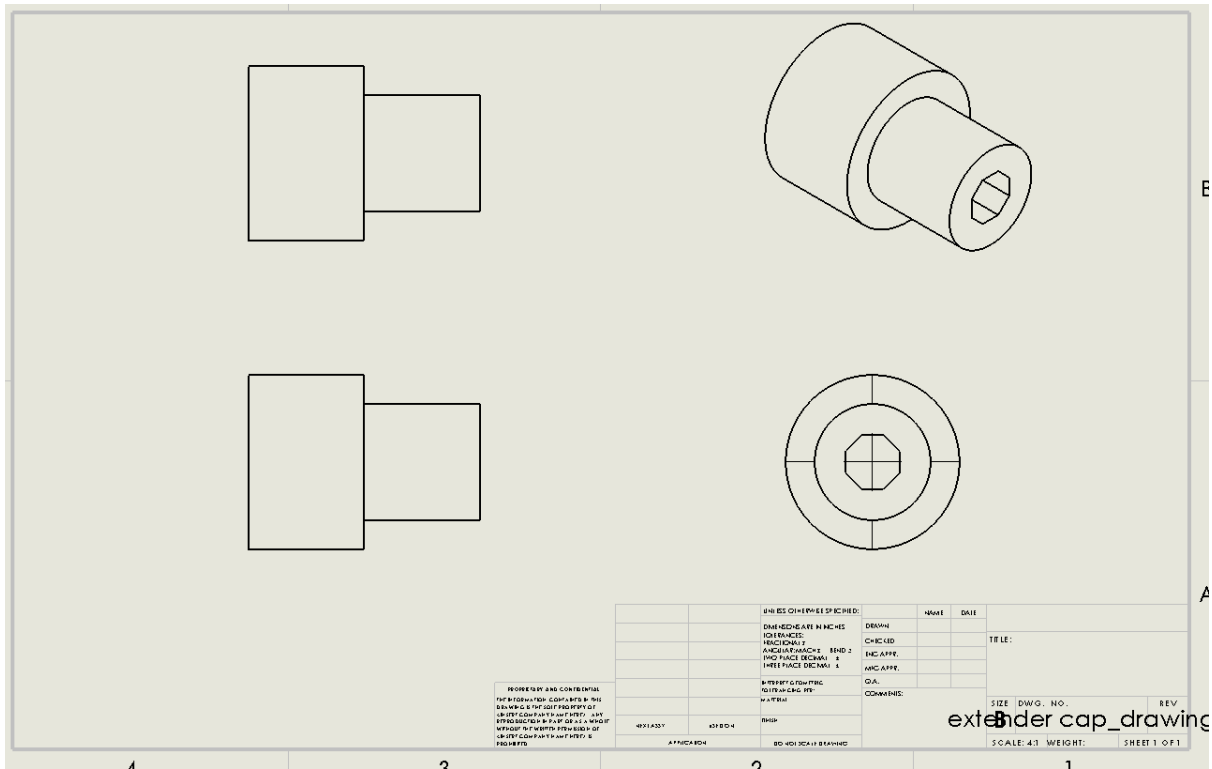


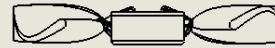
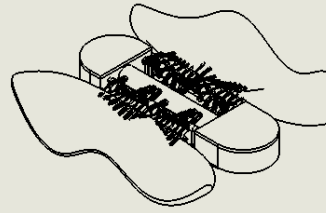
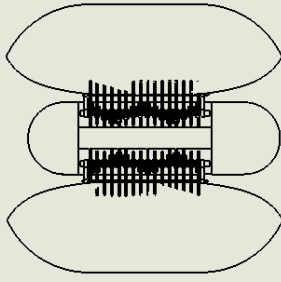


motionbar_drawing



external_cap_2_drawing





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Annexure 2

Annexure 2(a)

List of Structure components: like beams, bushes, shafts, belts, plates, pins, pulleys, wheel, connectors, batteries, motors, etc

- 1. Battery(1)**
- 2. Motors(2)**
- 3. Connectors(1-2)**

Annexure 2(b)

List of Motion Components: like Chain, sprockets, flaps, etc.

- 1. Fins**
- 2. Crank shaft**
- 3. Actuator**
- 4. Links**

Annexure 2(c)

List of Electronics Components: like Smart ports, switches, joysticks, controllers, LED/LCD screen, power supply, programming components, etc.

- 1. Arduino Nano**
- 2. Raspberry pi**
- 3. FlySky RC controller**
- 4. IMU**
- 5. PCB (Printed Circuit Board)**
- 6. Motor Driver**
- 7. L298N motor driver**
- 8. Raspberry pi camera**
- 9. Connecting Wires**

Annexure 2(d)

List of Other Accessories: Clothes, plastic eyes/ear/feeling like real all external components which are for the look

1.Chassis - This we will 3d print on based of our dimensions

2. Plastic Filament - PLA. Similar to PCL, PLA is the most commonly used **plastic filament** in desktop 3D printers; hence, it is readily available and is already being used in FDA-approved sutures like Vicryl (Ethicon Inc).

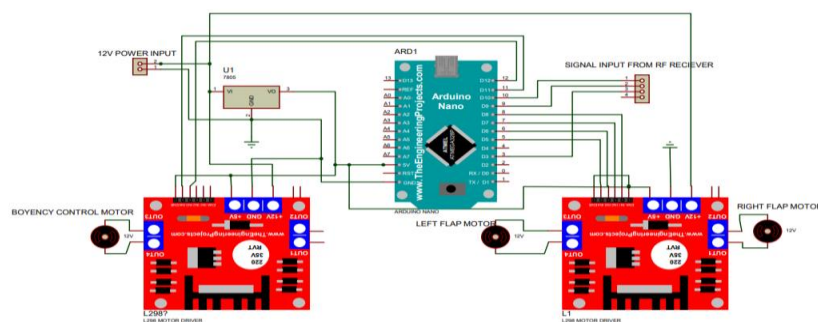
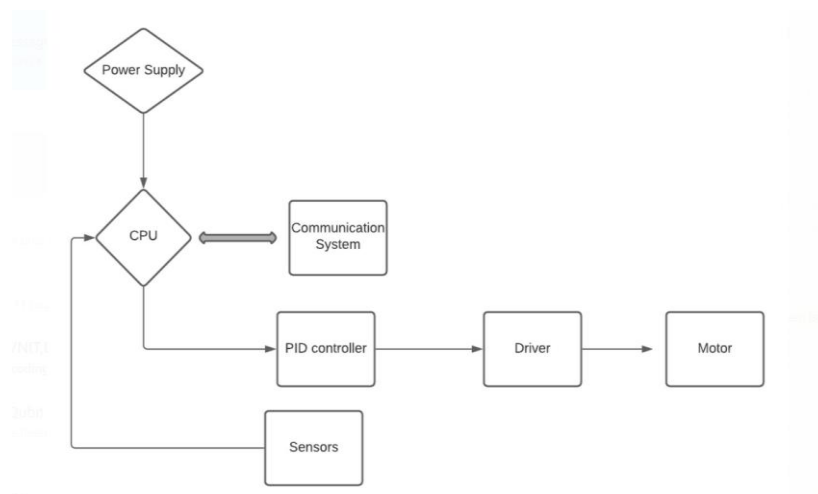
Annexure 3

The methodology of Making Robot: (Please write Entire Technical Specifications of Proposed Robot with brief notes and diagrams)

Proof of concept - We are going to make the complete body 3D printed including its chassis, shafts, links, etc. Then, we are going to assemble them as per design. To show its motion, we will just do it manually without any use of electrical components. In that way, We are going to show its functionality.

For Actual version -

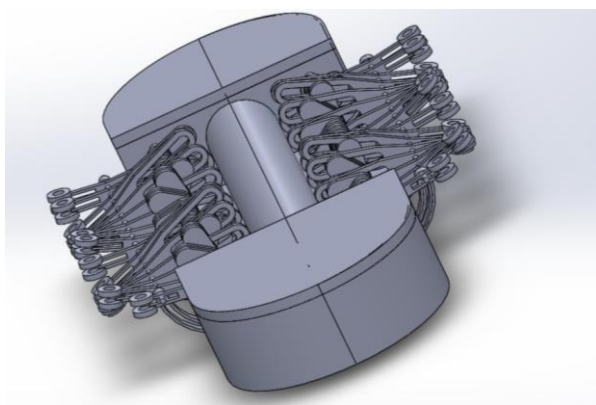
Our very first phase is to make a flow diagram of our whole robot in which it will include how the sensors, motors, controllers are going to interact with each other. Our basic flow diagram will look something like this below fig. And basic communication is going to be done using a Receiver - Transmitter wirelessly.



Now our next phase is going to design our basic model in Solidworks. We will also be using ANSYS for analysis and modify our design further to make it more efficient. Solidworks will be used for further updating in our design.

The mechanism on which our robot is going to swim underwater is based on the aquatic animal- Cuttlefish and its fins which run along its whole body.

Just like this cuttlefish which uses its fin to navigate underwater, we are going to have the same kind of streamlined body. In our design, the main shaft is made up of small links which are fixed inside slotted rods which in turn are connected with the fin. The same thing is present on both sides. The model will be similar to the image given below.



It is basically based on the concept of undulated fins which is going to help our robot to swim underwater and it can change its direction of propagation by changing the direction of rotation of one of the motors. By this, it can take **turns**.

In order to move it Up- Down , we are going to principle of force balancing i.e. between buoyant force and force due to gravity. For that we are going to use syringes which store water and then due to linear actuation (using a pneumatic actuator or other actuation), it will release some water that changes the weight of the body and In that way, we can adjust the depth of our robot (Just like how submarines works).

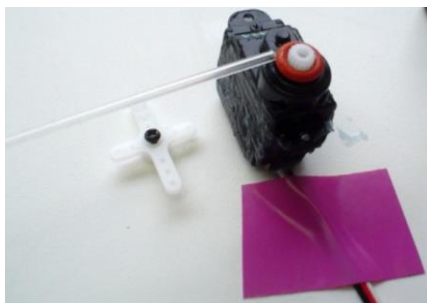


Most of the parts are going to be 3D printed(using **plastic filament**).We will try to make our whole design to be well optimised and effective and analyse it in all available ways before 3d printing it and will do adjustments or changes as per our needs.

After Printing all of them, We are going to assemble them, install all the electrical components to make it ready for testing and evaluation underwater.

Waterproofing will be done using **Oil Filled Servo** (old SoR method), the superglue and O-ring method.

Method - This is to fill the servo with a non-conductive oil. There are many types of oil we can use, such as cooking oil, mineral oil or silicone oil. We all decided to use light viscosity mineral oil because it's cheaply available. We are all thinking of using a heavy viscosity oil though, as it doesn't seep through small cracks in the seal. An illustration is shown in the below image.



For future aspects, We are planning to use a floating antenna from which our robot is tethered. So that for more depths say 40 ft , then it can go and do its application. And also we are planning to make this robot Amphibious means it can walk underwater as well as on land also.

Battery that we are going to use - Orange 5200mAh 3S 40C/80C Lithium polymer battery Pack (LiPo)

Specifications-

Capacity (mAh)	5200
Weight (gm)	360
Output Voltage (VDC)	11.1
Charge Rate (C)	1 ~ 3
Max. Charge Rate	5 C

For Motors-- Orange OG555 High Torque DC Motor 12V 500 RPM 45.8N-cm Encoder
Compatible

Specifications-

Input Voltage Range (DC)	6V to 15V
Rated RPM (at 12V)	450
Gear Reduction	20K
Rated Torque (N-cm)	45.8
Shaft Diameter (mm)	6mm

Connectors - XT60 Female Connectors-2pcs (Genuine)

Specifications-

Connector Type	XT60H
No. of Pins	2
Rated Current(A)	60
Rated Operating Voltage (VDC)	500
Weight	Â 2.4g

Annexure 4

5. Application of proposed Robot in a societal context: (Not more than 100 words)

- **Study of Aquatic life and habitat:** With their ability to withstand harsh environments, these bots can collect samples, observe marine life and test water temperatures.
- **Underwater Exploration** can make exploration cheaper and more accessible.
- Used to **check for maintenance of hulls of ships**(when integrated with AI algorithms) .
- Can be used by fishermen **to attract or observe fishes.**
Survey parts of water bodies and bring back samples for microplastic analysis and other environmental tests.